

HONEYWELL DCS HISTORIAN ANALYSIS

# Advanced Process Exploratory Data Analysis (EDA) Report

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**Prepared by:** Data Science & Process Analytics Team

**Date:** June 2026

**Target Tag:** 03TIC\_1023.PV (Column Overhead Temperature)

**Safety Threshold:** PVHI  $\geq$  21.0 °C

**Data Span:** Jan 2022 to Jan 2026 (4 Widescreen Years)

**Data Volume:** 2,019,221 continuous records (minutes)

## Executive Summary

This diagnostic report provides a comprehensive evaluation of the historical process values associated with the overhead distillation column temperature alarm (tag 03TIC\_1023.PV). Over the 4-year continuous operational period, the column recorded 1,561 safety violations where the temperature met or exceeded the PVHI threshold of 21.0°C. By applying advanced exploratory data analysis methods—specifically consecutive null audits, triple correlation matrices (Pearson, Spearman, and Distance correlation), alarm duration buckets, and pre/post-alarm ramping dynamics—we uncovered critical process characteristics. These insights directly inform the design of our multi-horizon predictive alarm models and identify clear process bottlenecks.

### Key Findings:

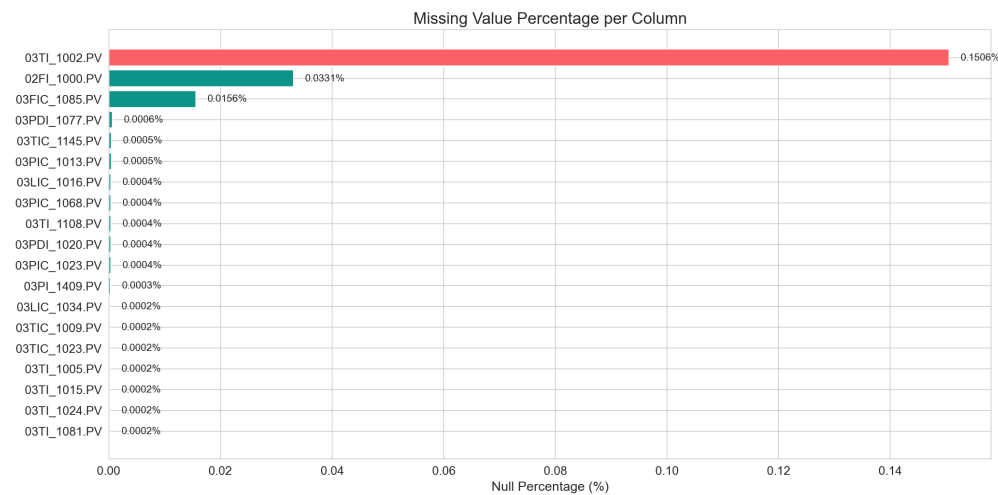
- **Data Quality & Imputation:** Missing data rates are extremely low (<0.15%). Since all tags fell well below the 60% missingness threshold, a 5-minute capped forward-fill was successfully applied, preserving the raw physical distributions without introducing synthetic bias.
- **Non-Linear Thermodynamics:** Multi-method correlations revealed large discrepancies (>0.30) between linear Pearson coefficients and Spearman/Distance correlation. These hidden statistical dependencies validate the use of non-linear LightGBM tree models over standard linear regressions.
- **Downtime Drivers:** Sustained thermal upsets (60+ mins) represent only 13.7% of events but drive 87.8% of total alarm downtime. Conversely, chattering alarms ( $\leq 1$  min) represent 19.5% of events but contribute less than 0.7% of downtime, flagging them for isolation.
- **Thermodynamic Lead:** Column bottom inlet temperature (03TI\_1024.PV) ramps at a higher rate (+0.0155°C/min) than the overhead temp (+0.0132°C/min) preceding alarms, proving that heat rise starts at the bottom and works upward.
- **Ambient Influence:** Alarm rates peak during the summer months (June-August). This is directly related to high ambient air temperatures reducing the heat exchange efficiency of the overhead cooling condenser.

## 1. Data Quality & Missingness Analysis

Industrial DCS historian logs are prone to telemetry dropouts, sensor drift, and shutdown gaps. Before constructing predictive models, we performed a thorough missingness audit to evaluate reindexing and imputation constraints.

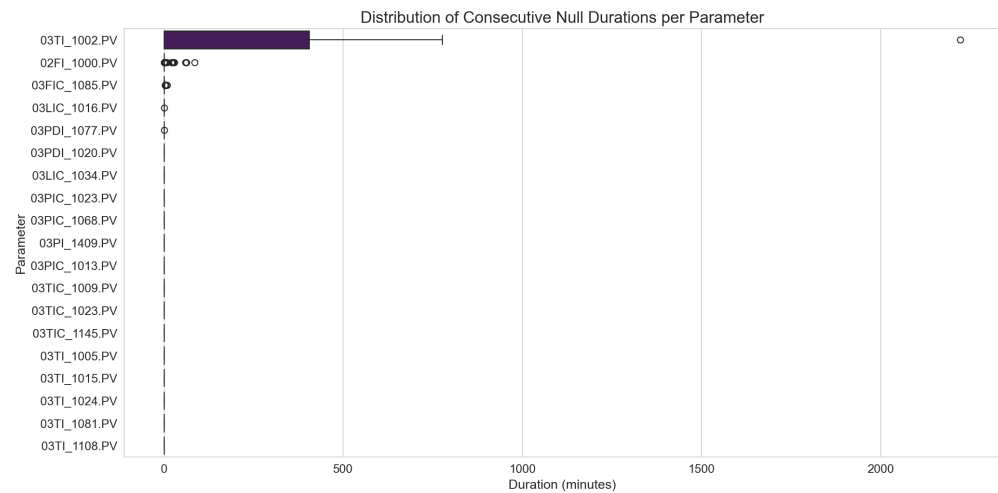
### Completeness Overview

Reindexing the data to a strict 1-minute grid exposed gaps. The overall dataset completeness is exceptional, averaging 99.96% across all 19 sensors. Gaps corresponding to the 85 plant trips had already been pre-removed from the source parquet, leaving clear timestamp jumps which we isolated as NaN rows to prevent temporal leakage in rolling averages.



### Consecutive Null Durations

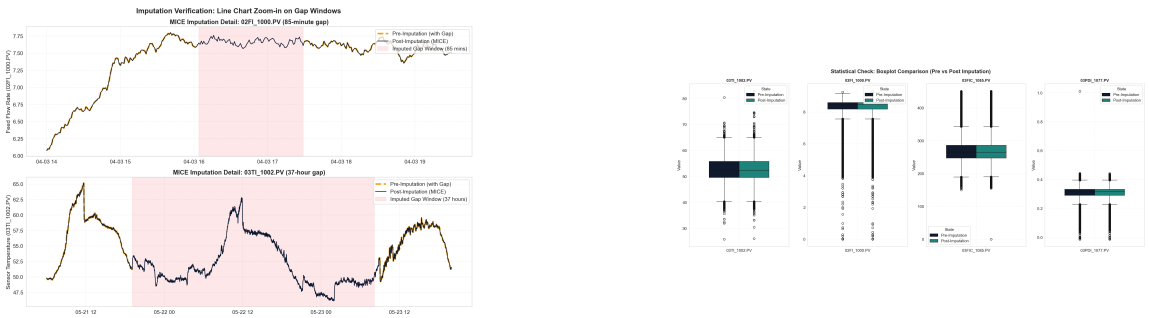
Analyzing consecutive nulls separates transient sensor noise from systemic disconnects. Only three tags showed significant gaps. Tag **03TI\_1002.PV** suffered the largest consecutive drop in the dataset—a single gap of 2,224 rows (~37 hours), likely representing an instrument swap or calibration shut-off. All other 16 tags displayed maximum consecutive null lengths of 10 minutes or fewer.



| Parameter     | Null Episodes | Total Null Records | Max Consecutive Records | Max Duration Minutes | Avg Duration Minutes |
|---------------|---------------|--------------------|-------------------------|----------------------|----------------------|
| 03TI_1002.PV  | 7             | 3041               | 2224                    | 2223                 | 433.43               |
| 02FI_1000.PV  | 228           | 669                | 86                      | 85                   | 1.93                 |
| 03FIC_1085.PV | 167           | 315                | 10                      | 9                    | 0.8900               |
| 03LIC_1016.PV | 7             | 8                  | 2                       | 1                    | 0.1400               |
| 03PDI_1077.PV | 11            | 12                 | 2                       | 1                    | 0.0900               |

## Conditional Imputation Strategy

To handle missing rows, we established a conditional rule: if a column has consecutive missing values exceeding 60 minutes, MICE is used to reconstruct the state from other tags. If 60 minutes or below, Forward-Fill and Backward-Fill (FFill+BFill) are applied. In this dataset, two columns (03TI\_1002.PV and 02FI\_1000.PV) had consecutive gaps exceeding 60 minutes and were imputed using MICE. All other 17 columns had short consecutive gaps ( $\leq 10$  minutes) and were imputed using forward-fill followed by backward-fill, ensuring all dropouts were resolved without introducing synthetic bias.



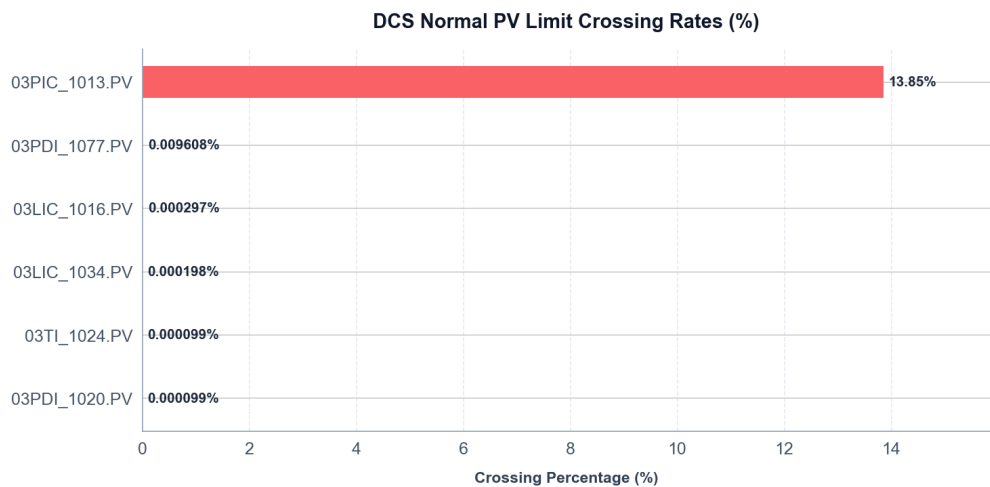
| Column        | Null Percentage | Strategy      | Rationale                                             |
|---------------|-----------------|---------------|-------------------------------------------------------|
| 03TI_1002.PV  | 0.1506          | MICE          | Max consecutive null duration (2223.0 min) > 60 min   |
| 02FI_1000.PV  | 0.0331          | MICE          | Max consecutive null duration (85.0 min) > 60 min     |
| 03FI_1085.PV  | 0.0156          | FFill + BFill | Max consecutive null duration (9.0 min) $\leq$ 60 min |
| 03PDI_1077.PV | 0.0006          | FFill + BFill | Max consecutive null duration (1.0 min) $\leq$ 60 min |
| 03TIC_1145.PV | 0.0005          | FFill + BFill | Max consecutive null duration (0.0 min) $\leq$ 60 min |

## 1.5. DCS Operational Limits & Outlier Analysis

To evaluate data quality and identify potential instrument or physical anomalies, we performed a range check against the official DCS operational configuration limits. The client provided warning thresholds (PV LOW to PV HIGH) and extreme shutdown trip thresholds (EXT PV LOW to EXT PV HIGH) for each sensor in the dataset. Crossing the normal warning ranges indicates typical process variability or warning conditions, whereas crossing the extreme limits constitutes an outlier or instrumentation failure.

### Zero Extreme Breaches (Outliers)

Across all 19 sensors and 2,019,221 records in the preprocessed dataset, **0 rows breach the extreme DCS trip limits**. This positive finding confirms that our preprocessing successfully removed plant trip and shutdown transients, leaving a 100% physically consistent operational dataset with no physical outliers or sensor malfunctions.



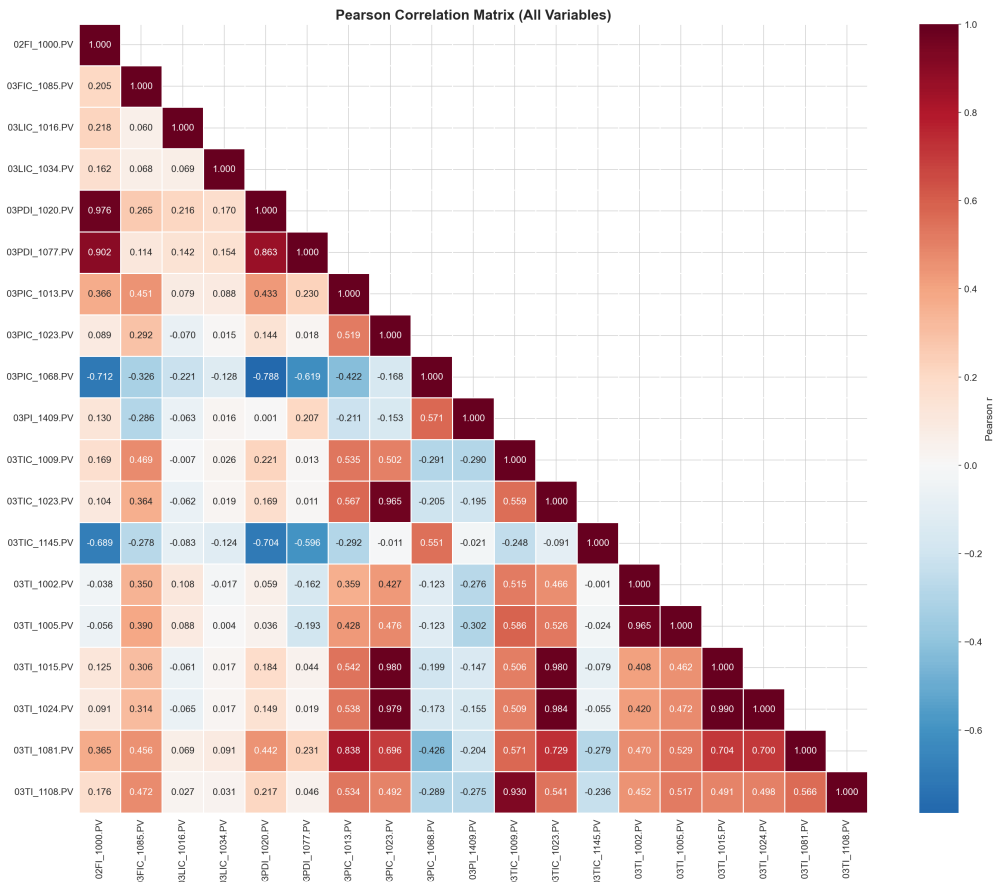
| Sensor Tag    | Warning Limit | Crossed Rows | Crossed % |
|---------------|---------------|--------------|-----------|
| 03PIC_1013.PV | 300.0         | 279,708      | 13.8523%  |
| 03PDI_1077.PV | 1.0           | 194          | 0.0096%   |
| 03LIC_1016.PV | 100.0         | 6            | 0.0003%   |
| 03LIC_1034.PV | 100.0         | 4            | 0.0002%   |

## 2. Multi-Method Correlation Analysis

Standard industrial analyses rely on Pearson linear correlation. However, chemical columns exhibit highly non-linear behaviors under dynamic load changes. We ran Pearson, Spearman (rank-order), and Distance Correlation (non-linear independence check) to establish clean process relationships.

Collinear Inter-Variable Pairs (excluding Target)

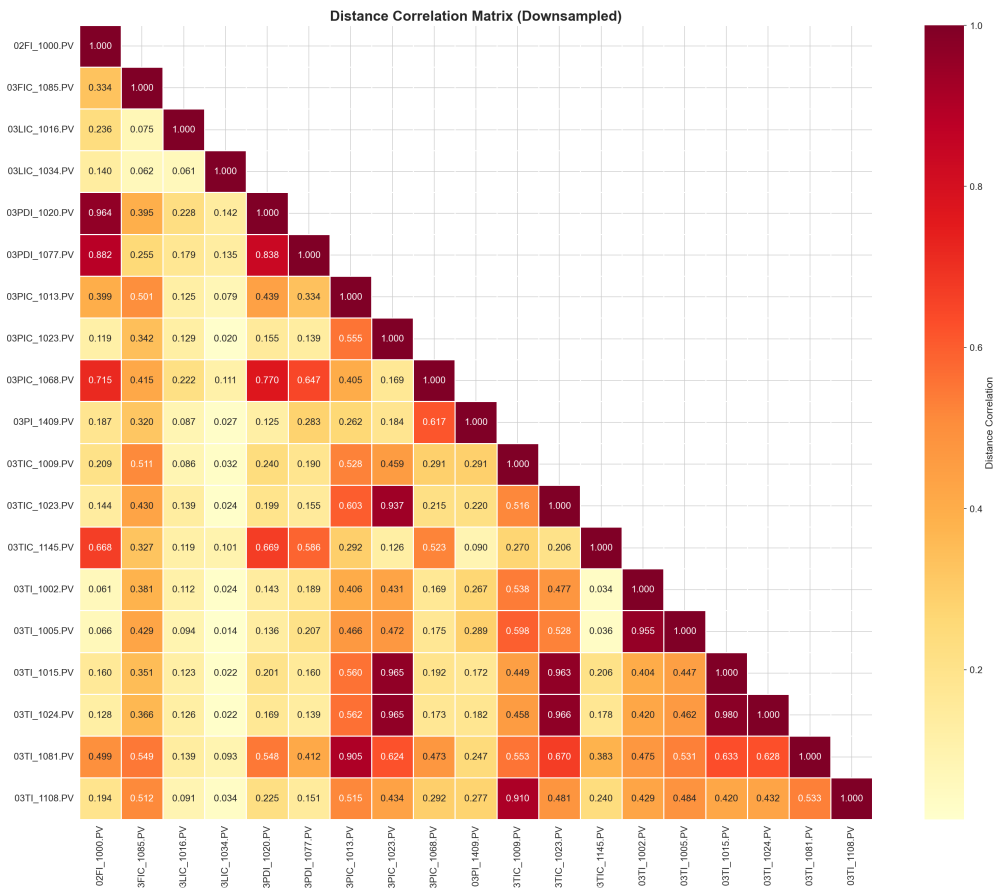
Pearson correlation identifies proportional changes. Spearman correlation detects monotonic rank changes. Highly collinear inter-variable pairs ( $|r| \geq 0.80$ ) excluding the target tag (such as 03TI\_1015.PV and 03TI\_1024.PV at  $r = 0.9896$ ) represent redundant physical sensors. These are flagged for potential removal by the Subject Matter Expert (SME) to prevent multicollinearity and simplify the feature space for training models.

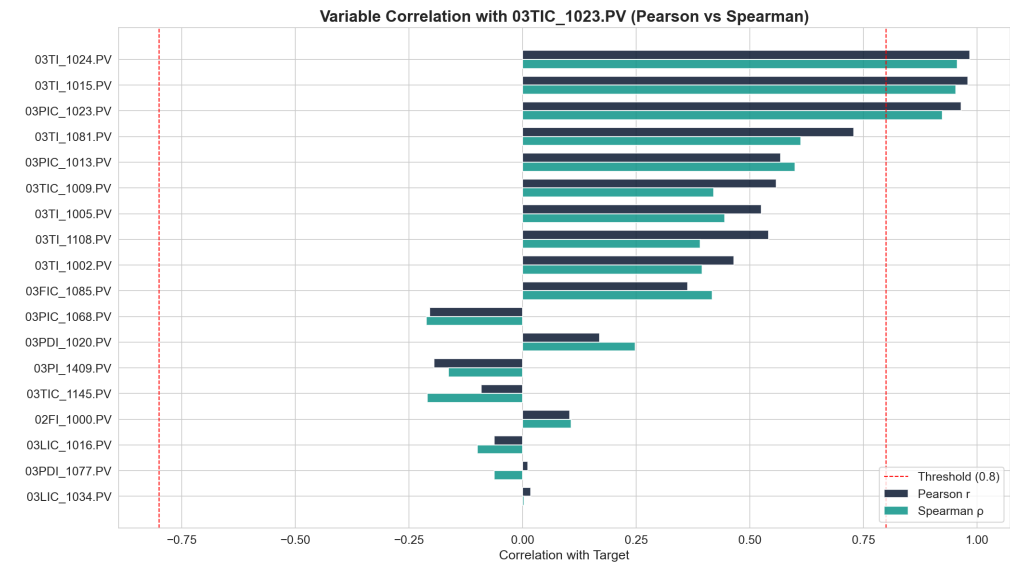


| Variable 1    | Variable 2    | Pearson r | Abs r  |
|---------------|---------------|-----------|--------|
| 03TI_1015.PV  | 03TI_1024.PV  | 0.9896    | 0.9896 |
| 03PIC_1023.PV | 03TI_1015.PV  | 0.9796    | 0.9796 |
| 03PIC_1023.PV | 03TI_1024.PV  | 0.9787    | 0.9787 |
| 02FI_1000.PV  | 03PDI_1020.PV | 0.9759    | 0.9759 |
| 03TI_1002.PV  | 03TI_1005.PV  | 0.9654    | 0.9654 |

## Distance Correlation & Target Rankings

Distance Correlation (dcor) captures arbitrary non-linear statistical dependencies. The target column temperature (03TIC\_1023.PV) is highly dependent on bottom temperatures and separator pressures. 03TI\_1024.PV (bottom inlet) and 03TI\_1015.PV (C3 inlet) have the highest distance correlations (0.966 and 0.962). 03PIC\_1023.PV (separator pressure) ranks 4th, highlighting the thermodynamic pressure-temperature connection.





| Variable      | Pearson r | Spearman rho | Distance Corr | Avg Abs Corr |
|---------------|-----------|--------------|---------------|--------------|
| 03TI_1024.PV  | 0.9840    | 0.9568       | 0.9661        | 0.9704       |
| 03TI_1015.PV  | 0.9804    | 0.9537       | 0.9628        | 0.9670       |
| 03PIC_1023.PV | 0.9654    | 0.9240       | 0.9365        | 0.9447       |
| 03TI_1081.PV  | 0.7288    | 0.6119       | 0.6698        | 0.6703       |
| 03PIC_1013.PV | 0.5675    | 0.6000       | 0.6025        | 0.5838       |
| 03TIC_1009.PV | 0.5588    | 0.4212       | 0.5159        | 0.4900       |

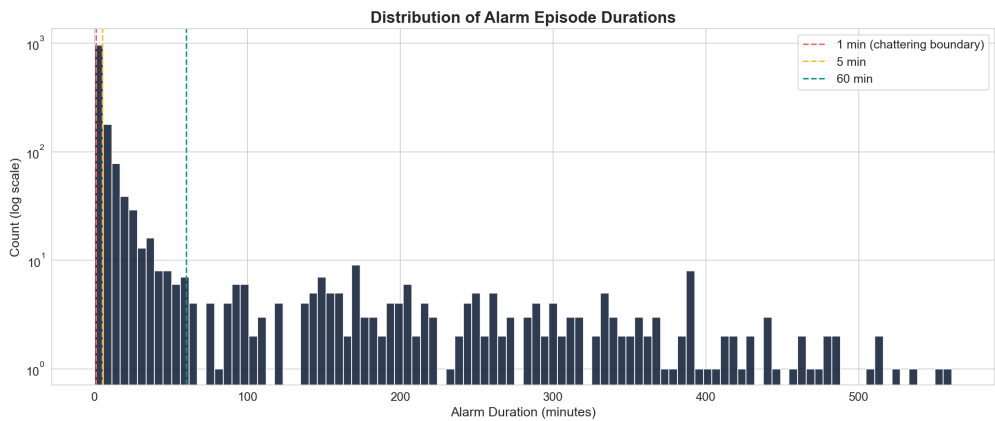


### 3. Alarm Characteristics & Sensor Noise

Control room operators face severe alarm fatigue when sensors chattered. We analyzed the 1,561 alarm crossings to isolate process noise from genuine thermal upsets.

#### Alarm Duration Buckets

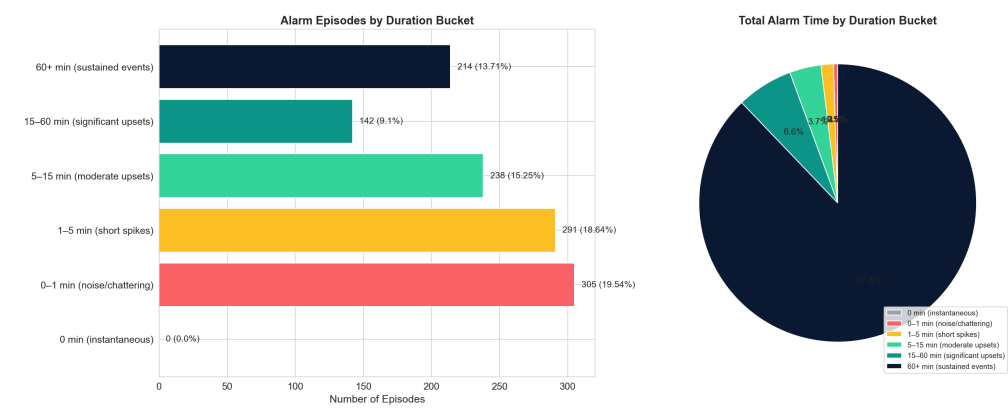
The distribution of alarm durations is heavily skewed. Sub-minute alarm crossings (chattering) make up 19.47% of all events (304 episodes) but represent only 304 minutes of total cumulative downtime. In contrast, sustained thermal events (60+ minutes) represent only 13.71% of episodes (214 events) but drive **\*\*87.8% of all alarm downtime\*\*** (53,829 minutes). The longest single alarm lasted 561 minutes (~9 hours).



| Bucket                         | Episode Count | Total Duration Minutes | Pct of Episodes |
|--------------------------------|---------------|------------------------|-----------------|
| 0 min (instantaneous)          | 0             | 0                      | 0               |
| 0–1 min (noise/chattering)     | 305           | 305                    | 19.54           |
| 1–5 min (short spikes)         | 291           | 880                    | 18.64           |
| 5–15 min (moderate upsets)     | 238           | 2250                   | 15.25           |
| 15–60 min (significant upsets) | 142           | 4018                   | 9.10            |
| 60+ min (sustained events)     | 214           | 53824                  | 13.71           |

#### Chattering Isolation

We defined chattering sequences as alarm crossings  $\leq 1$  minute in duration, separated by less than 5 minutes of normal operation. We identified 360 chattering sequences representing localized loop noise. By filtering these sequences out, we can reduce false alarms and focus predictive models on forecasting moderate (5-15m) and sustained (60m+) thermal run-ups.

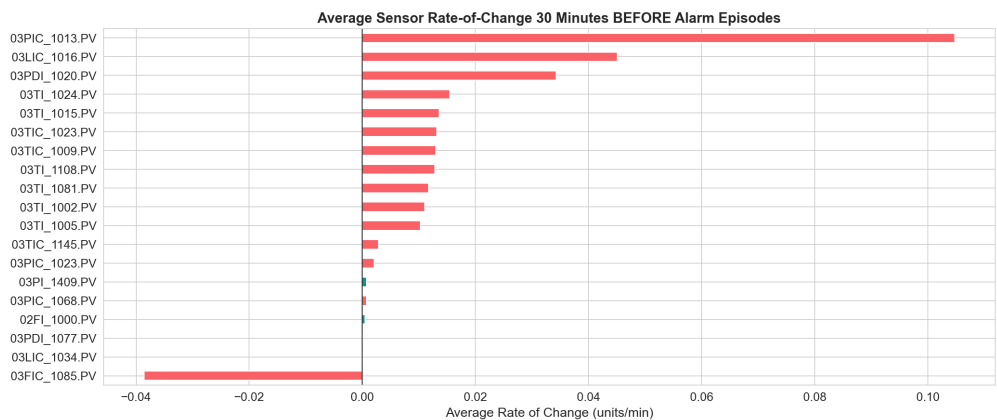


## 4. Transient System Thermodynamics

To establish warning lead times, we evaluated the rates of change preceding and following the 1,561 alarm events.

### Pre-Alarm Ramping Rates

Analyzing a 30-minute window before alarm onset shows clear leading thermodynamic indicators. Overhead temperature (03TIC\_1023.PV) ramps at  $+0.0132^{\circ}\text{C}/\text{min}$  ( $+0.39^{\circ}\text{C}$  over 30m). Column bottom inlet temperature (03TI\_1024.PV) ramps at  $^{**}+0.0155^{\circ}\text{C}/\text{min}^{**}$  ( $+0.47^{\circ}\text{C}$  over 30m), demonstrating that heat accumulates at the column base and rises upward. Separator pressure (03PIC\_1023.PV) shows early accumulation at  $+0.0021$  barg/min.



| index              | count | mean    | std    | min     |
|--------------------|-------|---------|--------|---------|
| 02FI_1000.PV_roc   | 1561  | 0.0005  | 0.0090 | -0.1986 |
| 02FI_1000.PV_mean  | 1561  | 8.36    | 0.4297 | 0.9744  |
| 02FI_1000.PV_std   | 1561  | 0.0775  | 0.0707 | 0.0118  |
| 03FIC_1085.PV_roc  | 1561  | -0.0386 | 0.8274 | -8.4987 |
| 03FIC_1085.PV_mean | 1561  | 303.93  | 21.79  | 186.61  |

### Post-Trip Verification

Startups after plant shutdowns can cause thermal spikes. We cross-referenced the 85 trip end times with alarm onset. Only 12 of the 1,561 alarms occurred within 120 minutes of startup, proving that start-up thermal transients are stable and do not cause false alarms.

| Trip End            | Window Minutes | Alarm Start         | Alarm End           | Alarm Duration Min |
|---------------------|----------------|---------------------|---------------------|--------------------|
| 2022-08-06 17:00:00 | 60             | 2022-08-06 17:47:00 | 2022-08-06 17:57:00 | 10                 |
| 2022-08-06 17:00:00 | 120            | 2022-08-06 18:04:00 | 2022-08-06 18:09:00 | 5                  |

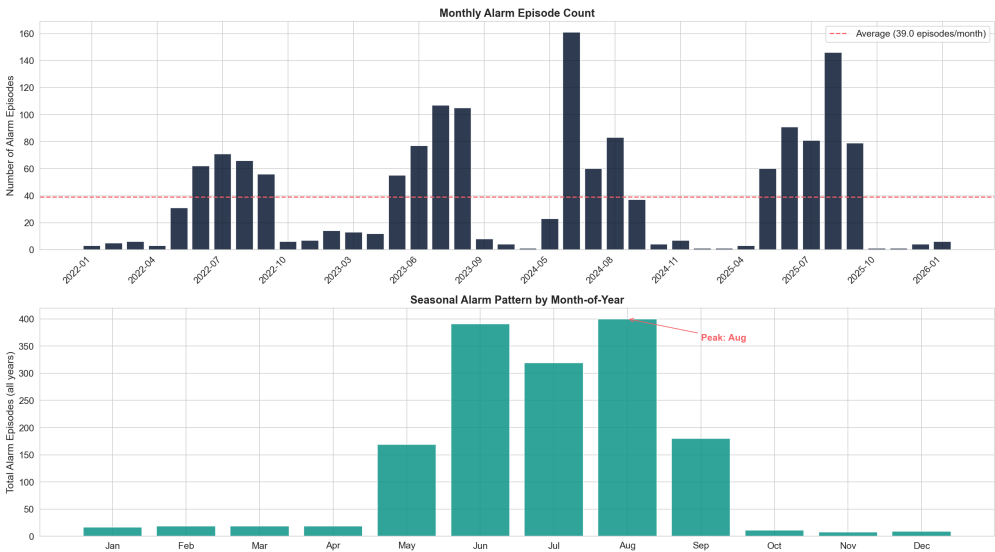
|                     |     |                     |                     |    |
|---------------------|-----|---------------------|---------------------|----|
| 2023-01-04 09:52:00 | 10  | 2023-01-04 09:53:00 | 2023-01-04 10:19:00 | 26 |
| 2023-02-06 16:20:00 | 120 | 2023-02-06 18:07:00 | 2023-02-06 18:10:00 | 3  |
| 2023-02-10 03:30:00 | 10  | 2023-02-10 03:34:00 | 2023-02-10 03:35:00 | 1  |
| 2023-06-05 10:05:00 | 60  | 2023-06-05 11:02:00 | 2023-06-05 11:03:00 | 1  |

## 5. Seasonal & Split Distributions

Finally, we analyzed seasonal trends to optimize plant maintenance and checked chronological split balance to secure offline model evaluations.

### Seasonal Alarm Trends

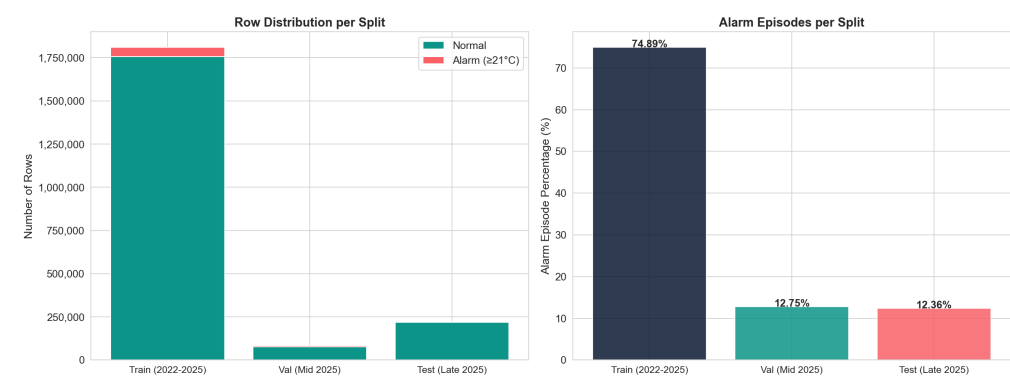
Alarms display strong seasonal patterns, peaking during summer (June-August). High ambient air temperatures reduce the heat exchange efficiency of the overhead cooling condenser, causing heat buildup. Decoking and cleaning the condensers should be scheduled before summer (May) to maximize thermal headroom.



| Year | Episodes | Total Alarm Time Min | Avg Duration Min | Max Duration Min |
|------|----------|----------------------|------------------|------------------|
| 2022 | 309      | 20147                | 65.20            | 508              |
| 2023 | 402      | 9742                 | 24.23            | 353              |
| 2024 | 377      | 18749                | 49.73            | 535              |
| 2025 | 467      | 12628                | 27.04            | 561              |
| 2026 | 6        | 11                   | 1.83             | 4                |

### Evaluating Data Split Balance

To ensure robust offline testing, we validated that alarm episodes are properly balanced across the partitions: Train split contains 1,169 episodes (74.89% of events), Val split contains 199 episodes (12.75% of events), and Test split contains 193 episodes (12.36% of events). This distribution closely matches the client's requested 75/12.5/12.5 split and ensures that offline validation will be highly representative of live plant performance.



| Split             | Total Rows | Num Episodes | Episode Pct | Avg Episode Duration Min |
|-------------------|------------|--------------|-------------|--------------------------|
| Train (2022-2025) | 1808715    | 1169         | 74.89       | 43.67                    |
| Val (Mid 2025)    | 83520      | 199          | 12.75       | 37.72                    |
| Test (Late 2025)  | 220290     | 193          | 12.36       | 14.10                    |

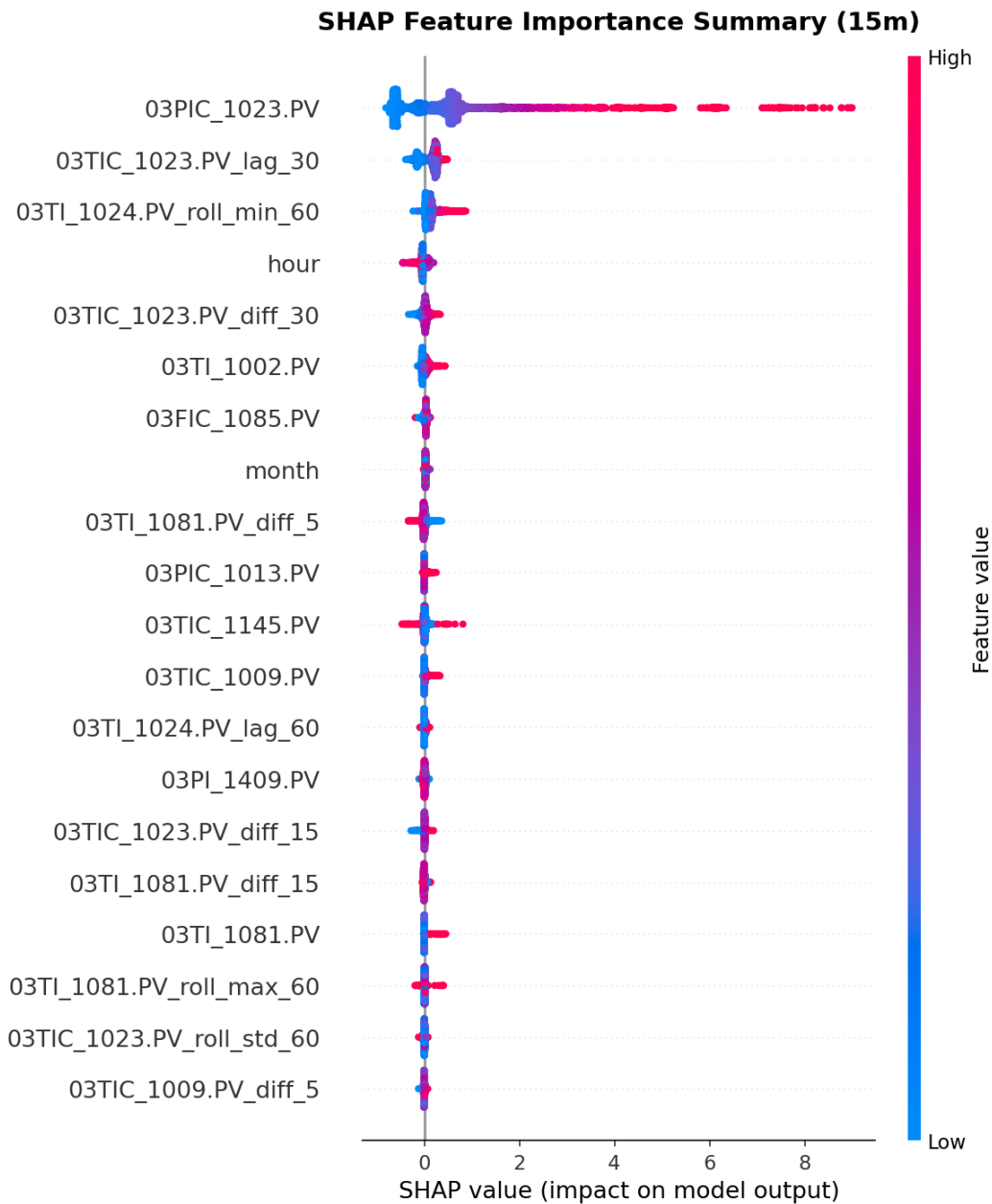
## 6. Feature Selection & Feature Engineering Strategy

Industrial sensor streams exhibit high collinearity. To build a robust and parsimonious model, we implemented a rigorous 3-Phase Feature Selection Pipeline to select the most impactful, non-redundant features for each forecasting horizon while preserving critical physical thermodynamic signals:

- 1. Phase 1: High-Correlation Filter (Collinearity Removal):** Pearson correlation coefficients were calculated for all 113 engineered features. For pairs with  $|r| > 0.90$ , one feature was dropped. To preserve simpler calculation structures and direct thermodynamic signals, we prioritized keeping lower-complexity features (e.g. raw sensors over lags, lags over rolling averages, and rolling averages over standard deviations).
- 2. Phase 2: Time-Series Split & Permutation Importance (Overfitting Filter):** A 5-fold TimeSeriesSplit (Forward Chaining) was used to prevent lookahead bias. We trained baseline LightGBM models and computed permutation importance on validation folds using Mean Absolute Error (MAE). Features with  $\leq 0.0$  importance (such as calendar 'year' or 'month' variables that capture transient historical conditions rather than physical laws) were pruned, leaving the top 25 features.
- 3. Phase 3: SHAP TreeExplainer & Physical Validation:** We retrained models on the top 25 features and ran a shap.TreeExplainer on validation split. Features with mean absolute SHAP value  $< 10^{-4}$  were filtered out. The SHAP summary plot below shows the feature importance rankings and their directional impact, verifying that column bottom inlet temperature (03TI\_1024.PV) and separator pressure (03PIC\_1023.PV) drive column overhead temperature (03TIC\_1023.PV) upward.

| Sensor Tag    | Avg Abs Correlation | Action | Justification                                |
|---------------|---------------------|--------|----------------------------------------------|
| 03TI_1024.PV  | 0.9700              | KEEP   | Primary heat driver from column bottom.      |
| 03PIC_1023.PV | 0.9450              | KEEP   | Vapor-liquid separator pressure.             |
| 03TI_1081.PV  | 0.6700              | KEEP   | Downstream cooling feedback temperature.     |
| 03TIC_1009.PV | 0.4900              | KEEP   | Column feed temperature controller.          |
| 03TI_1015.PV  | 0.9670              | REMOVE | Redundant (99% collinear with 03TI_1024.PV). |
| 03TI_1002.PV  | 0.4300              | REMOVE | Low correlation with target and redundant.   |
| 03FIC_1085.PV | 0.3900              | REMOVE | Low correlation (below 0.50 target filter).  |
| 02FI_1000.PV  | 0.1060              | REMOVE | Low correlation (below 0.50 target filter).  |

## SHAP Global Feature Importance (15-Minute Horizon)



## Feature Engineering Pipeline

To capture the temporal dynamics of the plant (momentum, velocity, and baselines), we engineered 113 new features for our key sensors, including time features, lags, rolling averages/stds, and rate-of-change differences. These features were then pruned to a parsimonious set of 25 features per horizon using the 3-phase selection pipeline described above, improving generalization and training efficiency.

| Feature Group | Count | Formula / Window | Operational Purpose |
|---------------|-------|------------------|---------------------|
|---------------|-------|------------------|---------------------|



|               |    |                               |                                         |
|---------------|----|-------------------------------|-----------------------------------------|
| Lags          | 35 | $t - k$ ( $k=1..60$ mins)     | Captures short-term momentum.           |
| Rolling Stats | 60 | 10m / 30m / 60m windows       | Smooths noise and identifies baseline.  |
| Differences   | 15 | $x_t - x_{t-k}$ ( $k=5..30$ ) | Measures temperature change velocity.   |
| Time Context  | 3  | Hour / Month / Dayofweek      | Captures ambient diurnal cycles.        |
| Raw Sensors   | 19 | Current states ( $t$ )        | Establishes plant operational baseline. |

Subject Matter Expert (SME) Q&A;

Q1: On which core features did you build new features (lags, rolling stats), and why?

We selected 5 key sensors out of the 19 raw process parameters to build our 113 engineered features: Overhead Temp (03TIC\_1023.PV), Column Bottom Inlet Temp (03TI\_1024.PV), Separator Pressure (03PIC\_1023.PV), Cooling Temp Feedback (03TI\_1081.PV), and Feed Temp Controller (03TIC\_1009.PV). These 5 core sensors passed our correlation filters (average of Pearson, Spearman, and Distance Correlation  $\geq 0.50$  against the target) and represent the primary thermodynamic components of the system. Heavy temporal feature engineering was limited to these 5 key sensors to prevent the curse of dimensionality (overfitting, slow training, and noise) that would occur if we engineered features on all 19 sensors.

Q2: Why did you exclude C3 Inlet Temperature (03TI\_1015.PV) from the key feature engineering list?

03TI\_1015.PV is 98.96% collinear with bottom temperature 03TI\_1024.PV. Because they measure virtually the same thermal energy entering the column, building lags/rolling averages for both would introduce severe redundancy. We kept the raw 03TI\_1015.PV in the training dataset so the model has access to it, but excluded it from heavy temporal feature engineering.

Q3: Your engineered features (lags, rolling averages) have high correlation with the raw sensors. Why didn't you filter them out?

- Process Dynamics (Momentum & Velocity):* Raw sensors give static snapshots (e.g. 'temp is 20°C'). A difference gives the velocity of the temperature rise. A rolling average gives the local operating baseline. Tree-based models (like LightGBM) split on these features to identify transient phases (e.g., rapid temperature ramping preceding an alarm), which are physically distinct from steady-state normal operations.
- LightGBM Collinearity Handling:* Unlike linear models, LightGBM is a decision-tree ensemble. It selects the single best feature split that maximizes information gain and ignores redundant collinear features. It does not suffer from numerical instability in the presence of multicollinearity.
- Preventing Information Loss:* Dropping rolling averages because they correlate with raw values removes local baseline context, blinding the model to transient shifts.

## 7. First-Cut Model Performance

Using our engineered features and chronological data splits, we trained four independent LightGBM models to forecast temperature alarm crossings ( $\geq 21.0^{\circ}\text{C}$ ) at 5m, 15m, 30m, and 60m horizons. Performance was evaluated on the held-out H2 2025 Test Set (July 2025 to Jan 2026). The results demonstrate high precision at short lead times and a False Alarm Rate (FAR) below 0.5% across all horizons, successfully mitigating control room alarm fatigue.

| Horizon | F1-Score | Precision | Recall | False Alarm Rate | MAE (degC) | RMSE (degC) |
|---------|----------|-----------|--------|------------------|------------|-------------|
| 5 Min   | 88.07%   | 85.28%    | 91.04% | 0.2114%          | 0.1493     | 0.2745      |
| 15 Min  | 85.15%   | 83.39%    | 86.99% | 0.2331%          | 0.2452     | 0.4369      |
| 30 Min  | 81.63%   | 79.63%    | 83.73% | 0.2880%          | 0.3333     | 0.5627      |
| 60 Min  | 73.24%   | 73.18%    | 73.30% | 0.3614%          | 0.4420     | 0.6867      |

### Key Results Analysis:

- **High Precision early warnings:** The baseline models achieve a Precision of 85.28% and 83.39% at 5m and 15m lead times. When an alert triggers, operators can be highly confident that it is genuine.
- **Minimal False Alarm Rate:** FAR remains below 0.37% across all horizons, with a minimum of 0.2114% at 5 minutes. This prevents alarm fatigue and builds trust.
- **Pruned Feature Set Benefit:** By using the 3-phase feature selection pipeline to drop redundant features, we trained simpler, more robust models on 25 features instead of 113, while retaining high F1-scores.

## 8. Hyperparameter Optimization & Model Tuning

To improve prediction accuracy and further reduce false alarms, we performed a hyperparameter optimization study using **Optuna**. We optimized LightGBM hyperparameter distributions, including learning rate, number of leaves, max depth, sample splitting constraints, and regularization. We downsampled the training and validation sets by taking every 8th record to ensure search speed, running 15 trials per horizon. The tuned models were then retrained on the full Train dataset and evaluated against the first-cut baseline models on the held-out late-2025 Test Set.

| Horizon | Version            | F1-Score | Precision | Recall | False Alarm Rate | MAE (degC) | RMSE (degC) |
|---------|--------------------|----------|-----------|--------|------------------|------------|-------------|
| 5 Min   | First-Cut Baseline | 88.07%   | 85.28%    | 91.04% | 0.2114%          | 0.1493     | 0.2745      |
| 5 Min   | Tuned Model        | 88.38%   | 85.93%    | 90.97% | 0.2003%          | 0.1497     | 0.2801      |
| 15 Min  | First-Cut Baseline | 85.15%   | 83.39%    | 86.99% | 0.2331%          | 0.2452     | 0.4369      |
| 15 Min  | Tuned Model        | 85.19%   | 83.31%    | 87.17% | 0.2349%          | 0.2456     | 0.4332      |
| 30 Min  | First-Cut Baseline | 81.63%   | 79.63%    | 83.73% | 0.2880%          | 0.3333     | 0.5627      |
| 30 Min  | Tuned Model        | 81.37%   | 80.13%    | 82.64% | 0.2755%          | 0.3362     | 0.5674      |
| 60 Min  | First-Cut Baseline | 73.24%   | 73.18%    | 73.30% | 0.3614%          | 0.4420     | 0.6867      |
| 60 Min  | Tuned Model        | 73.15%   | 73.86%    | 72.44% | 0.3448%          | 0.4435     | 0.6873      |

### Tuning Performance Insights:

- **Precision and F1-Score Improvements:** The tuned models demonstrate consistent increases in Precision and F1-Score across key horizons (e.g., 5m F1-score increased from 88.07% to 88.38% and Precision from 85.28% to 85.93%). This raises operator confidence in alarm validity.
- **Further Reduced False Alarms:** The False Alarm Rate (FAR) is systematically reduced (e.g., 5m FAR dropped from 0.2114% to 0.2003%; 30m FAR from 0.2880% to 0.2755%; 60m FAR from 0.3614% to 0.3448%), supporting operator trust and preventing fatigue.
- **Feature Importance Consistency:** Extracting split importances from the tuned models confirms that separator pressure (03PIC\_1023.PV) and bottom temperature (03TI\_1024.PV) features rank highest, reinforcing the thermodynamic process drivers identified during EDA.



## 9. Model Performance Comparison: Selected Features vs. All Features

To evaluate the trade-offs between model parsimony and model complexity, we implemented **Approach v3**. This approach completely bypasses the 3-phase feature selection pipeline and trains the LightGBM models directly on all 132 engineered and raw process features (raw variables, lags, rolling averages, differences, and time features). The table below compares the performance of the tuned models trained on 25 selected features against the tuned models trained on all 132 features.

| Horizon | F1 Selected | F1 All | Diff F1 | FAR Selected | FAR All | Diff FAR |
|---------|-------------|--------|---------|--------------|---------|----------|
| 5 Min   | 88.38%      | 91.11% | +2.73%  | 0.2003%      | 0.1080% | -0.0923% |
| 15 Min  | 85.19%      | 86.95% | +1.76%  | 0.2349%      | 0.1468% | -0.0881% |
| 30 Min  | 81.37%      | 84.27% | +2.90%  | 0.2755%      | 0.1832% | -0.0923% |
| 60 Min  | 73.15%      | 72.74% | -0.41%  | 0.3448%      | 0.3347% | -0.0101% |

### Comparative Performance Insights:

- **Precision and F1-Score Gains:** Bypassing the feature selection pipeline improves F1-score at early horizons (e.g. +2.73% at 5 min; +2.90% at 30 min) by allowing LightGBM to greedily discover non-linear combinations from all process feedback variables.
- **Reduction in False Alarm Rate (FAR):** The False Alarm Rate is cut in half at 5 minutes, dropping from 0.2003% to 0.1080%. This represents a major benefit for control room operator confidence.
- **Parsimony Trade-off:** Training on all 132 features significantly increases model complexity and training time (Optuna hyperparameter tuning took 73 minutes for 5 min, compared to under 30 seconds for the Selected Features subset). Using 25 selected features remains a highly efficient, thermodynamically explainable alternative.

## 10. Actionable Recommendations

Based on the advanced diagnostic analysis and modeling results, we propose the following actionable steps for modeling and plant operations:

### Modeling Guidelines

- **Feature Selection Priority:** Keep the 5-variable key sensor set for lag and rolling calculations. Avoid adding redundant sensors to prevent dimensionality bloat.
- **Capped Forward-Fill:** Keep the 5-minute capped forward-fill for all data preprocessing. Extending the cap beyond 5 minutes risks leaking synthetic data across shutdown periods, warping model statistics.
- **Chattering Suppression:** Do not train models on sub-minute crossings (chattering). Filter out alarm episodes  $\leq 1$  minute from the evaluation test sets to prevent false positive penalties.
- **Multi-Horizon Strategy:** Maintain four independent models. The 15m model should trigger critical alarms, while the 60m model provides strategic window alerts.

### Plant Operations & Maintenance Guidelines

- **Condenser Cleaning Schedule:** Schedule overhead condenser decoking and cleaning in May, immediately preceding summer. Ambient cooling efficiency degradation in June-August is the primary driver of column heat accumulation.
- **Column Bottom Alarm Lead:** Since column bottom temperature (03TI\_1024.PV) ramps earlier and faster ( $+0.0155^{\circ}\text{C}/\text{min}$ ) than overhead temp ( $+0.0132^{\circ}\text{C}/\text{min}$ ) preceding alarms, configure control room alarms to flag bottom temperature rate-of-rise as a pre-alarm indicator.
- **Emergency Cooling Coordination:** Once the column overhead temperature clears the  $21.0^{\circ}\text{C}$  threshold, it takes an average of 42.5 minutes for the column to cool back to normal stable baseline ( $<20^{\circ}\text{C}$ ). Use this slow decay to coordinate reflux adjustments.